

Welfare Analysis and Efficiency

Lecture 5

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Recap from Last Time

Quick Review: Competitive Market Equilibrium

Short-run equilibrium:

$$Q^D(P^*) = Q^S(P^*)$$

Long-run equilibrium: Free entry $\rightarrow P = MC = \min AC$, zero economic profits

Welfare result: Competitive equilibrium maximizes $CS + PS$

Tax incidence: The more inelastic side bears more of the tax

Today: What happens when markets are interconnected?

Why Move Beyond Partial Equilibrium?

Partial equilibrium analyzes one market in isolation — useful but limited

Example: A tax on gasoline

- In the gas market: price rises, quantity falls, DWL
- But what about the car market? The labor market for oil workers? The market for public transit?

Changes in one market ripple through others. **General equilibrium** accounts for all these interconnections simultaneously.

General Equilibrium: Setup

A Simple Economy

Two goods: x and y

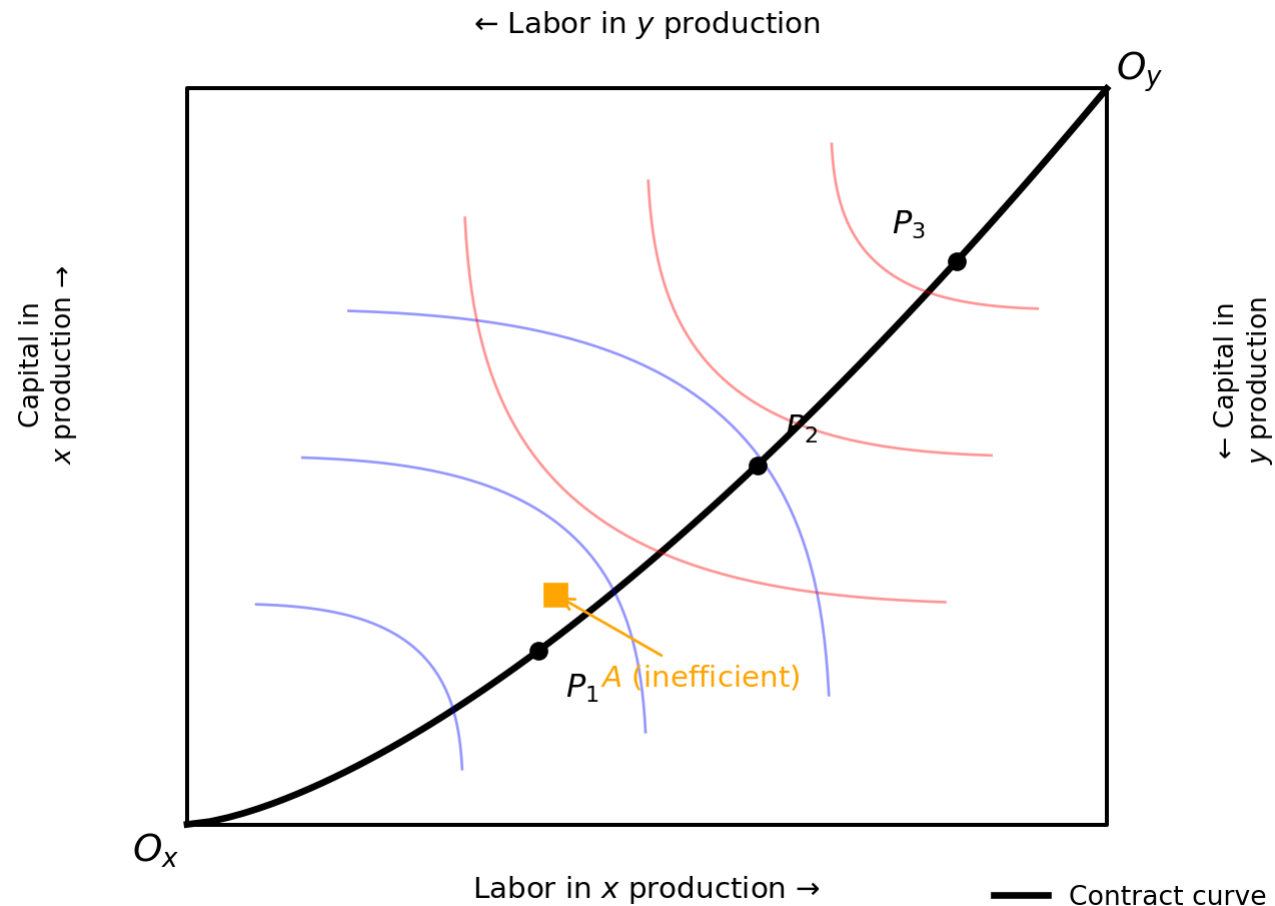
Two inputs: Labor (L) and Capital (K) — fixed total amounts $\bar{L}$ and $\bar{K}$

Firms: Produce x and y using L and K , maximize profits

Consumers: All have identical preferences, maximize utility

All markets are perfectly competitive: Everyone is a price-taker, law of one price holds, no transaction costs, perfect information

The Edgeworth Box for Production



Every point in the box is a full allocation of $\bar{L}$ and $\bar{K}$ between producing x and y. Efficient allocations lie on the **contract curve** where isoquants are tangent

Efficiency in Production

Efficient allocation requires:

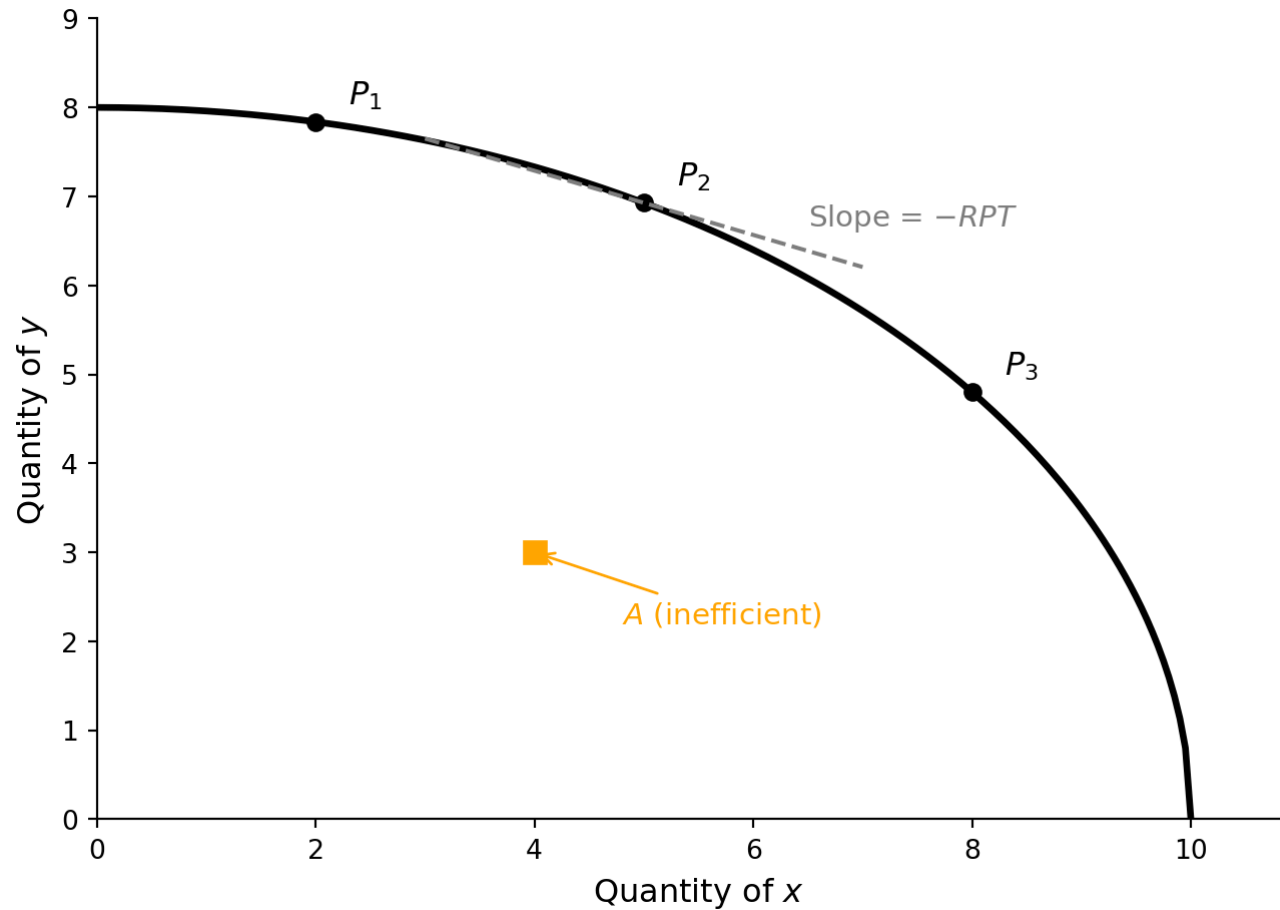
$$RTS_x(L \text{ for } K) = RTS_y(L \text{ for } K)$$

If the rates of technical substitution aren't equal, we can reallocate inputs to produce more of both goods

At point A in the previous figure: isoquants cross, so we can produce more X without sacrificing y (or vice versa) by moving to the contract curve

Production Possibility Frontier

The **contract curve** maps into the **production possibility frontier** (PPF):



Rate of Product Transformation

The **rate of product transformation (RPT)** is the negative of the slope of the PPF:

$$\text{RPT} = -\frac{dy}{dx} = \frac{MC_x}{MC_y}$$

Interpretation: The opportunity cost of producing one more unit of x in terms of foregone y

The PPF is concave (RPT increases) because:

- Diminishing returns: as we shift inputs toward x , marginal productivity of inputs in x falls
- Factor intensity differences: goods use inputs in different proportions

Deriving $RPT = MC_x / MC_y$

Set up the constrained maximization: for given $\bar{x}$, maximize y subject to resource constraints

Lagrangian:

$$\square = f_y(L_y, K_y) + \lambda_L(\bar{L} - L_x - L_y) + \lambda_K(\bar{K} - K_x - K_y) + \lambda_x(f_x(L_x, K_x) - \bar{x})$$

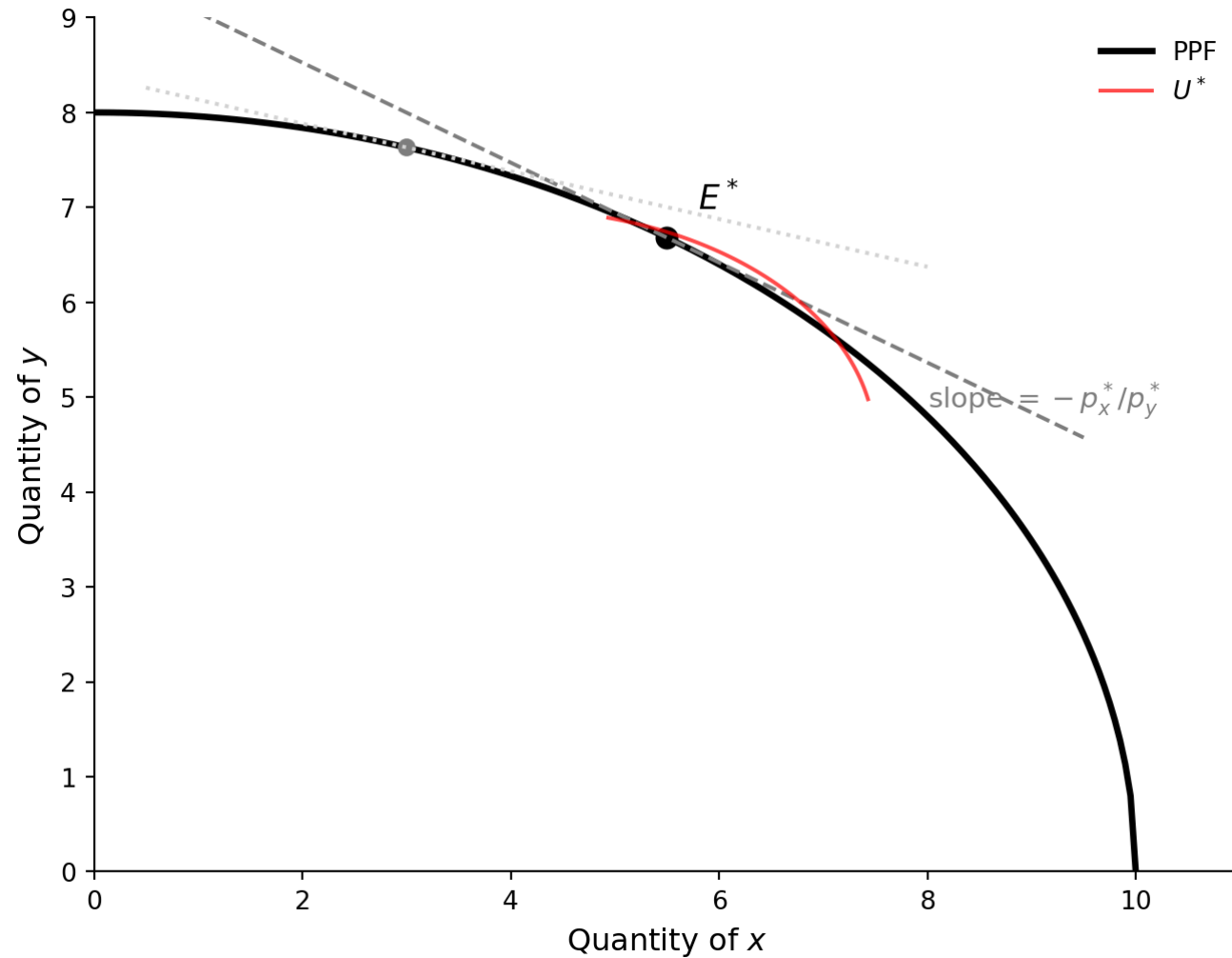
From the FOCs: $\lambda_L/\lambda_K = RTS$ in both industries (efficiency condition)

By the envelope theorem:

$$\frac{dy^*}{d\bar{x}} = -\lambda_x = -\frac{\lambda_L}{MP_L^x} = -\frac{MC_x}{MC_y}$$

Equilibrium Prices

Determination of General Equilibrium



Three Conditions for General Equilibrium

For the competitive equilibrium to hold simultaneously:

1. Profit maximization by firms:

$$RPT = \frac{MC_x}{MC_y} = \frac{p_x}{p_y}$$

2. Utility maximization by consumers:

$$MRS = \frac{p_x}{p_y}$$

3. Market clearing: Supply = demand for all goods and inputs

Together these imply: $RPT = MRS$ — the rate at which the economy *can* transform x into y equals the rate at which consumers are *willing* to trade x for y

Comparative Statics

Shift in preferences toward X:

- p_x/p_y increases
- Economy moves along the PPF: more X, less Y produced
- Both income and substitution effects at work

Technical progress in X production:

- PPF shifts outward (asymmetrically)
- p_x/p_y decreases
- More X consumed; effect on Y is ambiguous (income vs. substitution)

Application: The Corn Laws Debate

Historical context: Britain imposed high tariffs on grain after the Napoleonic wars (1815–1846)

Ricardo and the landlords:

- Tariffs kept grain prices high → more land devoted to grain
- Landowners (capital-intensive grain production) benefited
- Manufacturers wanted cheap grain so workers could accept lower wages

General equilibrium insight: Trade policy changes affect *all* factor prices, not just the product market. Opening trade shifts production along the PPF, changing the demand for capital vs. labor.

Application: Trade and Factor Prices

The **Stolper-Samuelson theorem** generalizes the Corn Laws logic:

- Opening trade increases the real return to the factor used intensively in the export sector
- Decreases the real return to the factor used intensively in the import-competing sector

U.S. implications:

- U.S. exports tend to be skilled-labor intensive
- Imports tend to be unskilled-labor intensive
- Trade liberalization increases the skill premium

This helps explain why unions of skilled workers tend to favor free trade while unions of unskilled workers tend to oppose it

Welfare Theorems

Pareto Efficiency

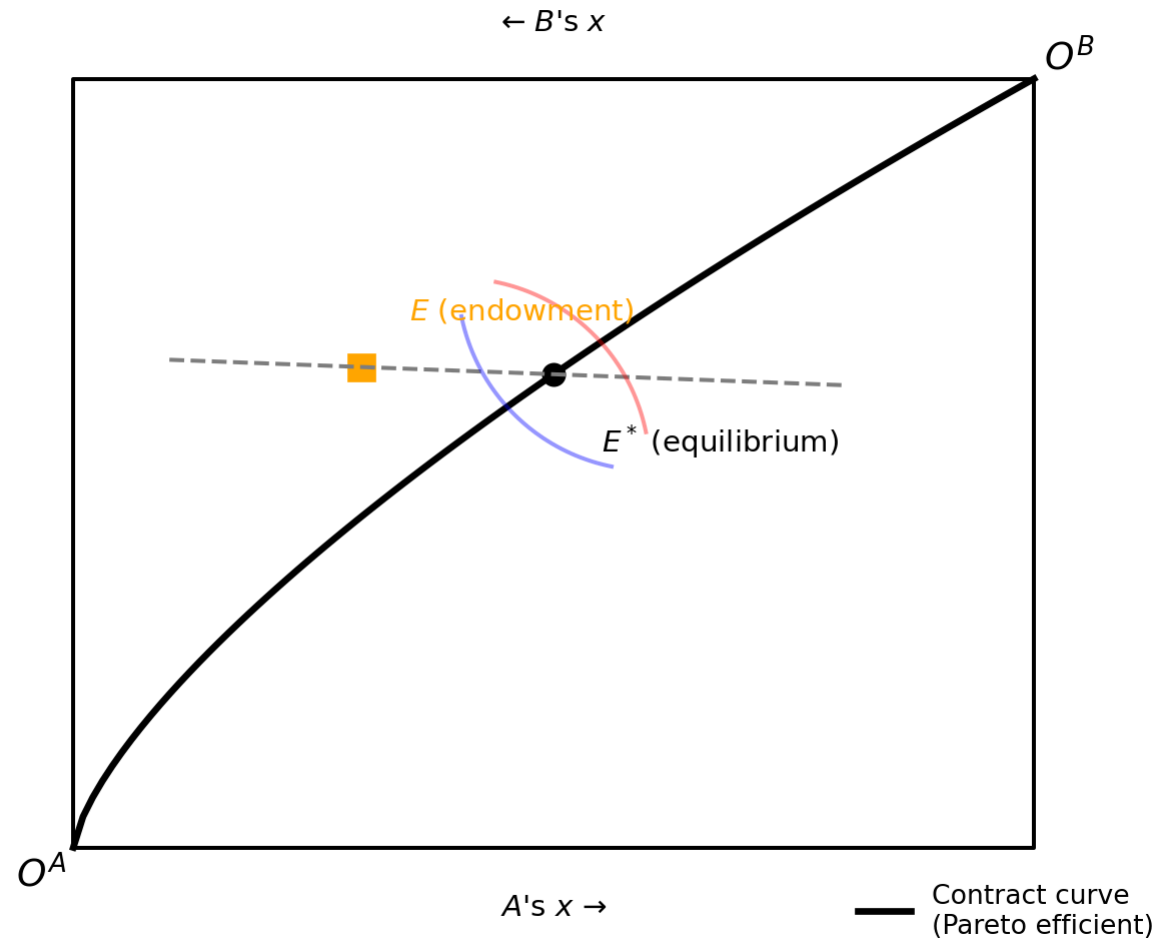
An allocation is **Pareto efficient** if there is no way to make someone better off without making someone else worse off

Three conditions for Pareto efficiency:

- **Exchange efficiency:** $MRS^A = MRS^B$ for all consumers A and B
- **Production efficiency:** $RTS_x = RTS_y$ for all inputs across industries
- **Output efficiency:** $RPT = MRS$ for all consumers

Competitive equilibrium satisfies all three!

The Edgeworth Box for Exchange



First Theorem of Welfare Economics

First Welfare Theorem: Every Walrasian (competitive) equilibrium is Pareto efficient.

Proof sketch: Suppose equilibrium allocation is *not* Pareto efficient. Then there exists a feasible reallocation making someone better off without hurting anyone. But if person i prefers bundle x_i' to their equilibrium bundle x_i^* , then x_i' must cost more than x_i^* at equilibrium prices (otherwise they would have chosen it). Summing over all individuals contradicts the feasibility constraint.

Powerful result: No central planner needed — competitive markets achieve efficiency on their own

What the First Welfare Theorem Does *Not* Say

The First Welfare Theorem says competitive markets are **efficient**, but:

- It says nothing about **equity** or **fairness**
- An allocation where one person has everything and others starve can be Pareto efficient
- Efficiency $\neq$ desirability

Also requires strong assumptions:

- Perfect competition (price-taking)
- Complete markets (no externalities, public goods)
- Perfect information
- No transaction costs

Second Theorem of Welfare Economics

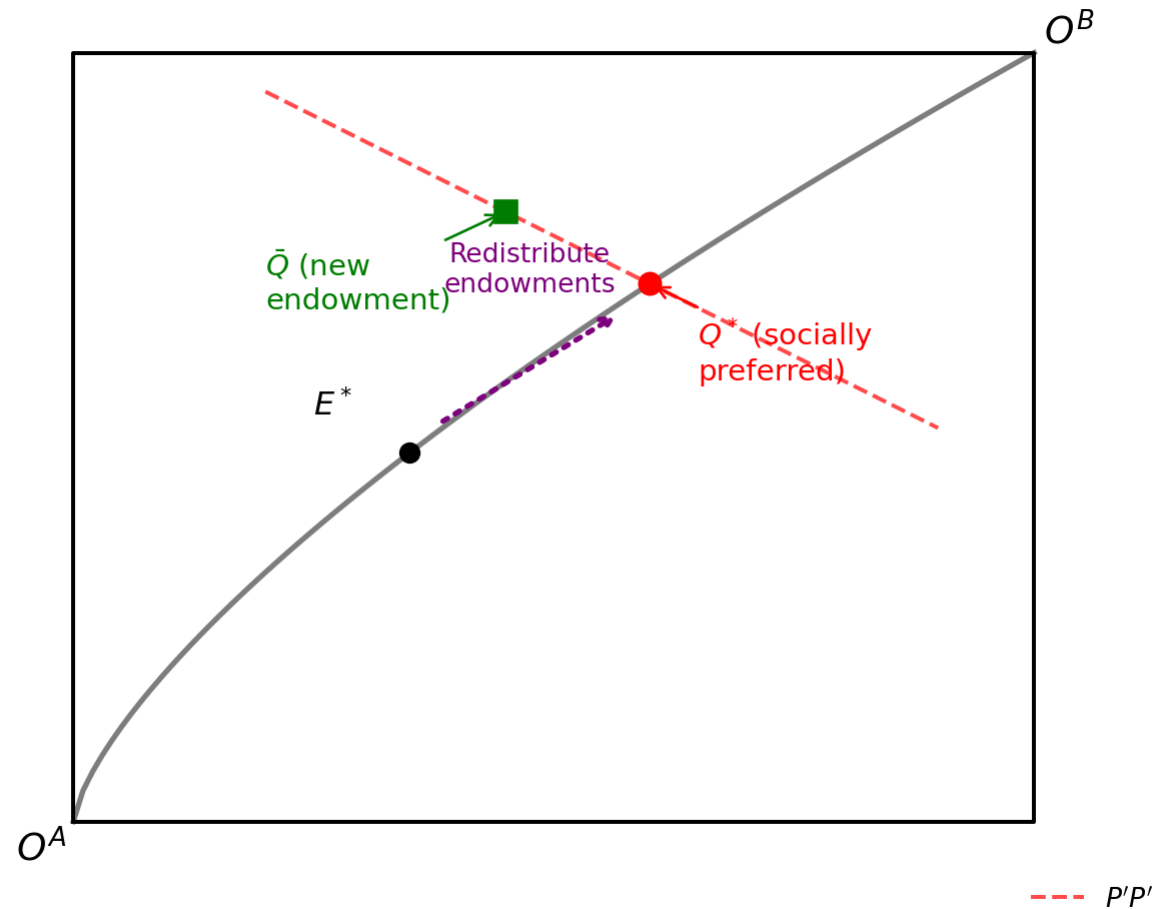
Second Welfare Theorem: For any Pareto efficient allocation, there exists a set of prices and a redistribution of initial endowments such that this allocation is a Walrasian equilibrium.

Implication: Efficiency and equity are *separable* problems

- Use lump-sum transfers to achieve desired distribution of endowments
- Then let markets work to achieve efficiency

In practice: Lump-sum transfers are difficult (require information about ability, not just income), so redistribution typically involves distortionary taxes → tradeoff between efficiency and equity

Visualizing the Second Welfare Theorem



Social Welfare

Social Welfare Functions

How do we choose among Pareto efficient allocations?

A **social welfare function** ranks allocations based on individuals' utilities:

$$SW = f(U_1, U_2, \dots, U_m)$$

CES form:

$$SW = \left[\sum_{i=1}^m U_i^R \right]^{1/R}, \quad R \leq 1$$

Special Cases of Social Welfare Functions

Utilitarian ($R = 1$):

$$SW = \sum_i U_i$$

Maximize total utility — treats a dollar the same regardless of who gets it

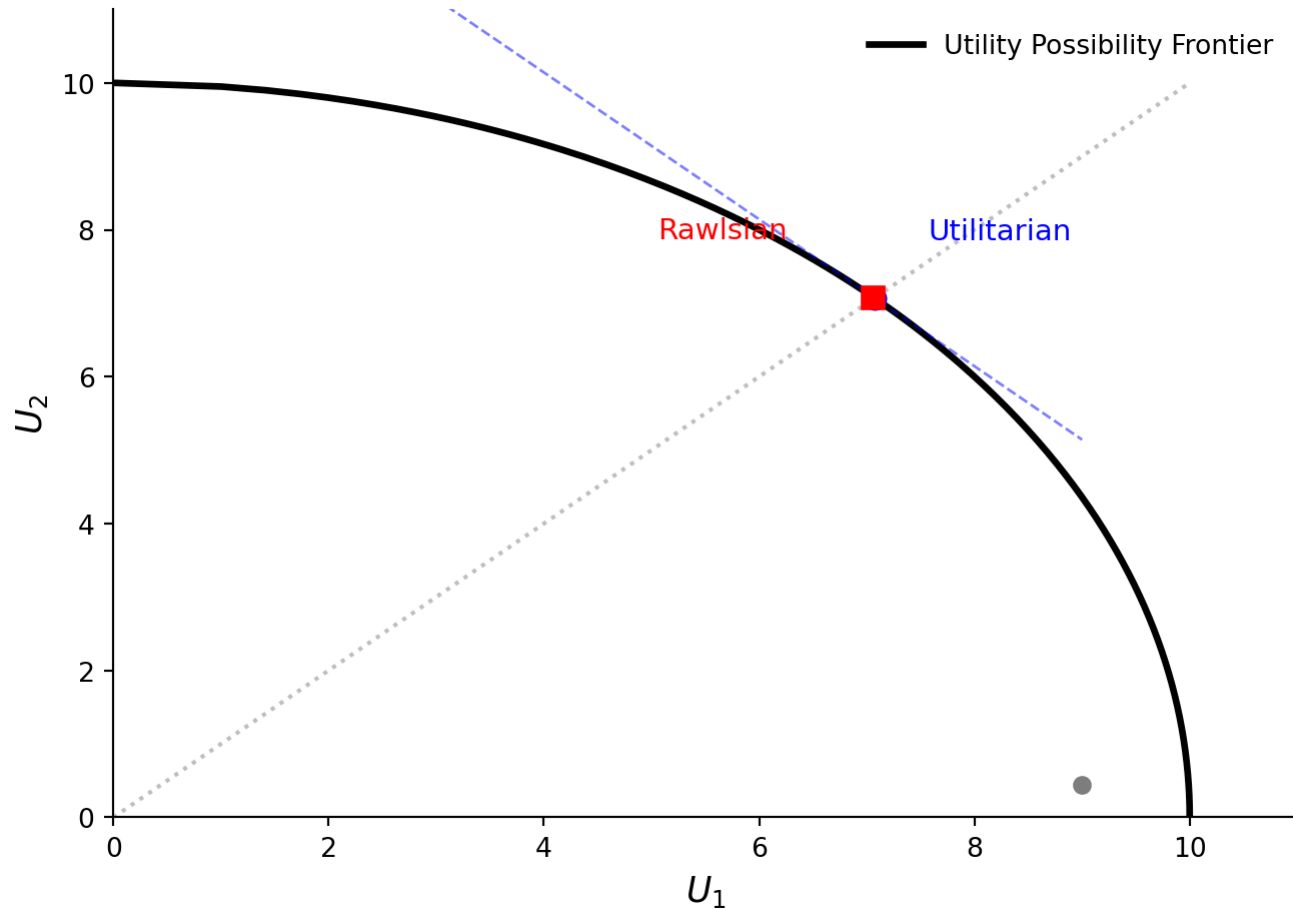
Rawlsian ($R \rightarrow -\infty$):

$$SW = \min_i \{U_i\}$$

Maximize the utility of the worst-off person — “maximin” criterion

Rawls’s argument: Behind a “veil of ignorance” (not knowing your position in society), risk-averse individuals would choose the maximin criterion

Choosing a Social Welfare Function



Every point on the utility possibility frontier is Pareto efficient. The social welfare function selects which efficient point society “should” be at.

Application: Evaluating Tax Reform

The welfare theorems provide a framework for policy evaluation:

Efficiency question (1st Welfare Theorem): Does the policy move us closer to or further from Pareto efficiency?

Equity question (2nd Welfare Theorem): Can we achieve the desired distribution through lump-sum transfers, or do we need distortionary policies?

In practice: Most policies involve tradeoffs between efficiency and equity because lump-sum transfers are infeasible. The debate is about how much inefficiency is worth accepting for more equitable outcomes.

Computable General Equilibrium

CGE Models: Taking Theory to Data

Computable general equilibrium (CGE) models apply the theoretical framework to real economies:

- Specify utility and production functions with estimated parameters
- Calibrate to actual economic data (national accounts, trade flows)
- Solve for equilibrium prices and quantities
- Simulate policy changes (tax reform, trade liberalization, environmental regulation)

Two key advances: Rich institutional detail + computational power

Applications of CGE Models

Trade policy: Nearly all CGE models found NAFTA offered welfare gains to all member countries, though distributional effects varied across sectors and skill groups

Tax policy: Simulate effects of tax reforms on labor supply, capital allocation, and income distribution. The Dutch MIMIC model emphasizes interactions with social welfare programs.

Environmental policy: Assess economic costs of emissions targets, compare carbon tax vs. cap-and-trade. Results depend heavily on how government revenue is recycled — a “double dividend” may arise if carbon tax revenue replaces distortionary income taxes

Application: Carbon Tax and General Equilibrium

Partial equilibrium view: Tax on carbon → higher energy prices → less pollution. Simple.

General equilibrium view: Much more complex:

- Higher energy costs raise production costs across all sectors
- Workers in carbon-intensive industries lose jobs
- Revenue can be recycled: cut income taxes? fund transfers? reduce deficit?
- International competitiveness effects (carbon leakage)

Optimal carbon tax depends on how environmental damage interacts with other forms of capital — this requires general equilibrium thinking

Try It Yourself

Consider an economy with two goods (x and y) and the production possibility frontier:

$$x^2 + y^2 = 100$$

Consumers have utility $U(x, y) = x \cdot y$

1. What is the RPT at a general point on the PPF?
2. Set up the conditions for general equilibrium: $RPT = MRS$
3. Solve for the equilibrium quantities of x and y
4. What is the equilibrium price ratio p_x/p_y ?

Summary

Key Takeaways

General equilibrium:

- All markets are interconnected — changes in one affect all others
- The Edgeworth box illustrates efficient input allocation
- The PPF shows the tradeoff between outputs; slope = $RPT = MC_x/MC_y$
- Equilibrium requires $RPT = MRS = p_x/p_y$ simultaneously

Key Takeaways (cont.)

Welfare theorems:

- **1st Welfare Theorem:** Competitive equilibria are Pareto efficient — markets “get it right” under ideal conditions
- **2nd Welfare Theorem:** Any efficient allocation can be decentralized through markets with appropriate redistribution
- **Implication:** Efficiency and equity are conceptually separable, though practically intertwined

Looking ahead:

- When assumptions of the welfare theorems fail → market failures
- Monopoly power, externalities, public goods, asymmetric information

Next Time

Lecture 6: Monopoly and Market Power

- Barriers to entry and sources of market power
- Monopoly pricing and output
- Deadweight loss of monopoly
- Price discrimination
- Regulation

Reading: Chapter 14